Joint Design of Beamforming and Reflection Coefficients in Integrated Sensing and Backscatter Communication Systems

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***Abstract*—Ambient Internet of Things networks leverage inexpensive and energy-efficient backscatter tags for a range of industrial uses. We propose an Integrated Sensing and Backscatter Communication (ISABC) system with multiple tags, a reader, and a full-duplex base station (BS) that performs both sensing and communication using the same resources. The BS can detect tags and communicate simultaneously by analyzing the backscattered signals from the tags, which also contain environmental data. We optimize the beamformers and tag reflection coefficients to reduce BS power consumption, using an alternating optimization method with closed-form solutions for the beamformers and semidefinite relaxation plus slack-optimization for other variables. In practice, ISABC can achieve a 75% improvement in combined communication and sensing rates compared to conventional systems, with only a 3.4% increase in transmit power, and when using active tags, the power increase is just 0.24% over standard sensing and communication methods.**

***Index Terms*—Backscatter communication (BackCom), Integrated sensing and communication (ISAC), Passive tags.**

I. Introduction

The future landscape of Internet-of-Things (IoT) networks is shaped by the necessity for low-power, high-quality wireless connectivity alongside precise and reliable sensing capabilities [1], [2]. The concept of ambient power-enabled (battery-free) Internet of Things stands out as a particularly active area of research in this evolving domain, attracting considerable interest to the extent that the 3GPP has initiated a dedicated study item [1]. Within this perspective, a groundbreaking concept introduced in [3] is known as *Integrated Sensing and Backscatter Communications (ISABC)*. This paradigm shift merges the principles of integrated sensing and communication (ISAC) with the capabilities of backscatter communication (BackCom), offering a solution that facilitates the concurrent execution of sensing and communication tasks in ambient power-enabled IoT networks.

On the one hand, ISAC offers a significant shift from traditional network models by allowing simultaneous sensing and communication. This approach diverges from the standard practice of using separate networks for each function, integrating these tasks into a cohesive system. This integration is crucial for the evolution from 5G to the realm of 6G technology [4]. Sensing within ISAC can be categorized based on the configuration of the transmitter and the sensing receiver — namely, mono-static, bi-static, and multi-static configurations — as well as by the type of sensing signal [4]. Signals are considered active when the sensing receiver utilizes the reflections or diffraction of its own transmissions, and passive when it uses signals transmitted by another source [4]. In active sensing scenarios, the receiver transmits and senses simultaneously, operating in a full-duplex (FD) mode [5]. A significant challenge in this mode is self-interference (SI), which can adversely affect sensing accuracy. In FD ISAC systems, SI cancellation is critical and must be carefully managed to eliminate direct signal coupling between transceiver antennas while retaining target reflections [5].

On the other hand, BackCom technology, particularly in the context of ambient-powered IoT networks, has garnered significant attention due to its energy and spectrum efficiency [6]–[9]. This technology utilizes passive tags that lack active RF components and communicate by reflecting external RF signals, i.e., backscattering. Such an approach addresses the challenges associated with battery-powered IoT devices, which typically necessitate regular battery replacements or recharging. These challenges include high maintenance costs, environmental issues, and, in certain scenarios, safety risks, such as with wireless sensors deployed in critical industries like power generation and petroleum.

While previous works study ISAC and BackCom systems separately [6], [7], [10], study [3] is the first one to introduce a holistic exploration of their integrated functionalities and ensuing performance metrics of a single user single tag ISABC system. This work extends on the initial work in [3] by investigating the multi-tag situation, breaking new ground by using their synergistic potential. To achieve minimal BS power consumption while adhering to the quality-of-service (QoS) requirements of each network node, we delve into optimizing BS transmit/receive beamformers and tag power reflection coefficients, despite the complexity introduced by the nonlinear energy harvesting (EH) model at the tags. We employ an alternative optimization (AO) strategy, beginning with the optimization of the BS's received beamforming via minimum mean-squared error (MMSE) filtering and the signal-to-noise-to-interference ratio (SINR), then determining the BS transmit beamforming through the semidefinite relaxation (SDR) method, and finally addressing the tag reflection coefficients through a slack-variable optimization approach. Through comprehensive convergence and complexity analysis, alongside simulations, we demonstrate ISABC's superiority in enhancing combined communication and sensing rates by 75% compared to traditional BackCom systems, with only a modest 3.4% increase in transmit power, thereby validating the effectiveness and efficiency of ISABC in modern network environments.

Notation: Vectors and matrices are expressed by boldface

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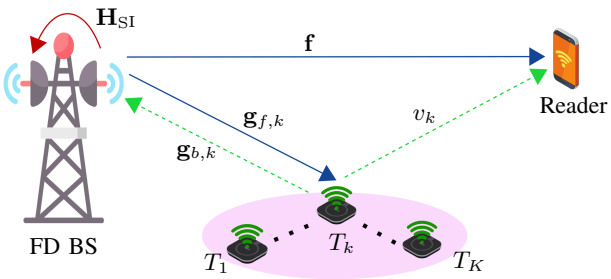


Fig. 1: An ISABC system setup.

lower case letters $\mathbf{a}$ and capital letters $\mathbf{A}$, respectively. For a square matrix $\mathbf{A}$, $\mathbf{A}^{\mathrm{H}}$ and $\mathbf{A}^{\mathrm{T}}$ are Hermitian conjugate transpose and transpose of a matrix, respectively. $\mathbf{I}_M$ denotes the M-by-M identity matrix. The Euclidean norm of a complex vector and the absolute value of a complex scalar are denoted by $\|\cdot\|$ and $|\cdot|$, respectively. The distribution of a circularly symmetric complex Gaussian (CSCG) random vector with mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{C}$ is denoted by $\sim \mathcal{CN}(\boldsymbol{\mu}, \mathbf{C})$. The expectation operator is denoted by $\mathbb{E}[\cdot]$. Besides, $\mathbb{C}^{M\times N}$ and $\mathbb{R}^{M\times 1}$ represent $M \times N$ dimensional complex matrices and $M \times 1$ dimensional real vectors, respectively. Further, $\mathcal{O}$ expresses the big-O notation. Finally, $\mathcal{K} \triangleq \{1, \ldots, K\}$ and $\mathcal{K}_k \triangleq \mathcal{K} \setminus \{k\}$.

II. System, Channel, and Signal Models

As shown in Fig. 1, we consider an ISABC network having an FD BS consisting of M transmit and N receiver uniform linear array (ULA) antennas, K single-antenna backscatter tags/sensors, denoted by T_k, for $k \in \mathcal{K}$, and a single-antenna user. The FD BS uses transmit beamforming for communication and environment sensing. Tags utilize the BS-transmitted signal to harvest energy and data transfer. A cooperative user decodes its data and then uses successive interference cancellation (SIC) for tag data. The BS also captures the tag-reflected signals to extract environmental insights [3].

We consider block flat-fading channel models for the system. During each fading block, the channels between the BS and the user, the BS and T_k, and T_k and the user are denoted by $\mathbf{f} \in \mathbb{C}^{M\times 1}$, $\mathbf{g}_{f,k} \in \mathbb{C}^{M\times 1}$, and $v_k \in \mathbb{C}$, respectively. Moreover, $\mathbf{g}_{b,k} \in \mathbb{C}^{N\times 1}$ represents the channel between T_k and the BS receiver antennas. Among these channels, pure communication channels, i.e., $\mathbf{f}$ and v_k, are modeled as Rayleigh fading and given by $\mathbf{a} = \zeta_a^{1/2}\tilde{\mathbf{a}}$, where $\mathbf{a} \in \{\mathbf{f}, v_k\}$. Here, ζ_a captures the large-scale path loss and shadowing, which stays constant for several coherence intervals, and $\tilde{\mathbf{a}} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_A)$ accounts for the small-scale Rayleigh fading[1], where $A \in \{M, 1\}$.

Conversely, following the echo signal multiple-input and multiple-output (MIMO) radar, the channels between the BS and tags are modeled as line-of-sight (LoS) paths [5]. We denote the transmit/receiver array steering vectors to the direction θ_k by

$$\mathbf{b}(\theta_k) = \sqrt{\frac{\zeta_b}{B}}\left[1, e^{j\pi\sin(\theta_k)}, \ldots, e^{j\pi(B-1)\sin(\theta_k)}\right]^{\mathrm{T}}, \tag{1}$$

[1]Note that $v_k = \zeta_{v_k}^{1/2}\tilde{v}_k$ and $\tilde{v}_k \sim \mathcal{CN}(0,1)$.

where $\mathbf{b} \in \{\mathbf{g}_{f,k}, \mathbf{g}_{b,k}\}$, $B \in \{M, N\}$, θ_k is the direction of T_k with respect to the BS and user/reader direction, and ζ_b is the path-loss. Finally, the SI channel between the transmitter and the receiver ULAs of the BS is denoted as $\mathbf{H}_{\mathrm{SI}} \in \mathbb{C}^{M\times N}$.

The BS transmitted signal $\mathbf{x} \in \mathbb{C}^{M\times 1}$, which includes both data and sensing waveforms, is given by $\mathbf{x} = \mathbf{w}x_d + \mathbf{s}$, where $x_d \in \mathbb{C}$ is the intended data symbol for the mobile user/reader with unit power, i.e., $\mathbb{E}\{|x_d|^2\} = 1$, $\mathbf{w} \in \mathbb{C}^{M\times 1}$ is the BS beamforming vector, and $\mathbf{s} \in \mathbb{C}^{M\times 1}$ is the sensing signal with the covariance matrix $\mathbf{S} \triangleq \mathbb{E}\{\mathbf{s}\mathbf{s}^{\mathrm{H}}\}$ for extending the degrees-of-freedom of $\mathbf{x}$ to achieve enhanced sensing performance [5]. Also, it is assumed that x_d and $\mathbf{s}$ are independent of each other, and the beamforming at the BS is achieved through designing $\mathbf{w}$ and $\mathbf{S}$ [5].

The user receives the BS signal as well as the tags' backscattered signals. The user-received signal is thus given by

$$y = \mathbf{f}^{\mathrm{H}}\mathbf{x} + \sum_{k\in\mathcal{K}} \sqrt{\alpha_k}\mathbf{h}_k^{\mathrm{H}}\mathbf{x}c_k + z_u, \tag{2}$$

where the first and the second terms in (2) represent the direct-link, i.e., BS-to-user, and backscatter-link, i.e., BS-to-tags-to-user, signals, respectively. Moreover, $z_u \sim \mathcal{CN}(0, \sigma^2)$ is the white Gaussian noise (AWGN) at the user with 0 mean and σ^2 variance, $\mathbf{h}_k$ is the effective backscatter channel through T_k, i.e., $\mathbf{h}_k = \mathbf{g}_{f,k}(\theta_k)v_k$, and c_k is T_k's data with $\mathbb{E}\{|c_k|^2\} = 1$. As the mobile user/reader and BS are integral to the primary networks, a pre-existing connection facilitates the exchange of information, including sensing waveform, via a control link. It is thus assumed that the user knows the sensing waveform in advance and removes it before decoding data. Following the removal of the sensing signal, the received signal can be expressed as

$$y = \mathbf{f}^{\mathrm{H}}\mathbf{w}x_d + \sum_{k\in\mathcal{K}} \sqrt{\alpha_k}\mathbf{h}_k^{\mathrm{H}}\left(\mathbf{w}x_d + \mathbf{s}\right)c_k + z_u. \tag{3}$$

Next, the user performs SIC to recover the backscattered data from the tags. In particular, the user decodes its own signal, treating tag signals as interference, and then subtracts the decoded x_d from the received signal (3) for decoding the tags' data. The post-processed signal for decoding tags' data is thus given as

$$y_t = \sum_{k\in\mathcal{K}} \sqrt{\alpha_k}\mathbf{h}_k^{\mathrm{H}}\left(\mathbf{w}x_d + \mathbf{s}\right)c_k + z_u. \tag{4}$$

Backscattered signals from tags, on the other hand, reach not only the user but also the BS. The BS thus aims to extract environmental information from these unintentionally received backscattered signals [3]. The received signal at the BS, i.e., $\mathbf{y}_b \in \mathbb{C}^{N\times 1}$, is given as

$$\mathbf{y}_b = \sum_{k\in\mathcal{K}} \sqrt{\alpha_k}\mathbf{G}_k(\theta_k)\mathbf{x}c_k + \mathbf{H}_{\mathrm{SI}}^{\mathrm{H}}\mathbf{x} + \mathbf{z}_b, \tag{5}$$

where $\mathbf{G}_k(\theta_k) \triangleq \mathbf{g}_{b,k}(\theta_k)\mathbf{g}_{f,k}^{\mathrm{H}}(\theta_k)$ and $\sqrt{\alpha_k}\mathbf{G}_k(\theta_k)\mathbf{x}c_k$ is the T_k's reflection. The second term in (5) denotes the SI at the receiver of the BS due to simultaneous transmission and reception, and $\mathbf{z}_b \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I}_N)$ the AWGN at the BS. We assume that the FD BS cancels the SI at its receiver using

perfect SI cancellation techniques [11]. The post-processed SI cancelled signal is thus given as

$$\mathbf{y}_b' = \sum_{k\in\mathcal{K}} \sqrt{\alpha_k}\mathbf{G}_k(\theta_k)\mathbf{x}c_k + \mathbf{z}_b. \tag{6}$$

The BS then applies the receiver beamformer, $\mathbf{u}_k \in \mathbb{C}^{N\times 1}$ for $k \in \mathcal{K}$, to the received signal (6) to capture the desired reflected signal of T_k. The post-processed signal for obtaining T_k's sensing information is given as

$$\begin{aligned} y_{b,k} &= \sqrt{\alpha_k}\mathbf{u}_k^{\mathrm{H}}\mathbf{G}_k(\theta_k)\mathbf{x}c_k \\ &\quad + \sum_{i\in\mathcal{K}_k} \sqrt{\alpha_i}\mathbf{u}_k^{\mathrm{H}}\mathbf{G}_i(\theta_i)\mathbf{x}c_i + \mathbf{u}_k^{\mathrm{H}}\mathbf{z}_b. \end{aligned} \tag{7}$$

The SINRs for sensing and communication tasks substantially impact the performance of both systems. The user first decodes its data, considering the tags' signal as interference. From (3), the received SINR is obtained as

$$\Gamma_u = \frac{|\mathbf{f}^{\mathrm{H}}\mathbf{w}|^2}{\sum_{k\in\mathcal{K}} \alpha_k(|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}|^2 + \mathbf{h}_k^{\mathrm{H}}\mathbf{S}\mathbf{h}_k) + \sigma^2}. \tag{8}$$

Here, we consider that while the user can cancel interference from the direct-link sensing signal, i.e., $\mathbf{f}^{\mathrm{H}}\mathbf{s}$, it cannot cancel interference from tag-sensing reflections because they consist of unknown tag data, i.e., c_k for $k \in \mathcal{K}$. The user employs the SIC for decoding the backscatter data. Using (4), the SINR of T_k at the user is given as

$$\Gamma_{t,k} = \frac{\alpha_k\left(|\mathbf{h}_k^{\mathrm{H}}\mathbf{w}|^2 + \mathbf{h}_k^{\mathrm{H}}\mathbf{S}\mathbf{h}_k\right)}{\sum_{i\in\mathcal{K}_k} \alpha_i\left(|\mathbf{h}_i^{\mathrm{H}}\mathbf{w}|^2 + \mathbf{h}_i^{\mathrm{H}}\mathbf{S}\mathbf{h}_i\right) + \sigma^2}. \tag{9}$$

The BS uses the unintentionally received backscattered signal for sensing, i.e., to learn and obtain environmental information. The BS applies a receiver beamformer, $\mathbf{u}_k$ for $k \in \mathcal{K}$, to the received signal (6) to capture the desired reflected signal of T_k. To this end, the sensing SINR of T_k is obtained using (7) which is given by

$$\Upsilon_k = \frac{\alpha_k\mathbf{u}_k^{\mathrm{H}}\mathbf{G}_k(\theta_k)\mathbf{R}_x\mathbf{G}_k^{\mathrm{H}}(\theta_k)\mathbf{u}_k}{\mathbf{u}_k^{\mathrm{H}}\left(\sum_{i\in\mathcal{K}_k}\alpha_i\mathbf{G}_i(\theta_i)\mathbf{R}_x\mathbf{G}_i^{\mathrm{H}} + \sigma^2\mathbf{I}_N\right)\mathbf{u}_k}, \tag{10}$$

where $\mathbf{R}_x \triangleq \mathbb{E}\{\mathbf{x}\mathbf{x}^{\mathrm{H}}\} = \mathbf{w}\mathbf{w}^{\mathrm{H}} + \mathbf{S}$ is the covariance matrix of the BS transmitted signal [5].

The tags in this system are passive, relying on EH instead of batteries, as they do not emit RF signals on their own. This is achieved through a power-splitting mechanism that divides the incoming RF power into two parts: one for data transmission (reflected power) and the other for EH. We use a parametric nonlinear sigmoid EH to model the total harvested power at T_k as $p_k^{\mathrm{h}} = \Phi((1-\alpha_k)p_k^{\mathrm{in}})$ [12], where

$$\Phi(p_k^{\mathrm{in}}) = \psi_k^{\mathrm{NL}} - \frac{M_{\mathrm{NL}}\Omega_{\mathrm{NL}}}{1-\Omega_{\mathrm{NL}}}, \quad \Omega_{\mathrm{NL}} = \frac{1}{1+\exp(a_{\mathrm{NL}}b_{\mathrm{NL}})}, \tag{11}$$

$$\psi_k^{\mathrm{NL}} = \frac{M_{\mathrm{NL}}}{1+\exp(-a_{\mathrm{NL}}(p_k^{\mathrm{in}} - b_{\mathrm{NL}}))}, \quad \forall k, \tag{12}$$

where ψ_k^{NL} is a standard logistic function with constant Ω_{NL} ensuring a zero input/output response. Parameters a_{NL} and b_{NL} represent circuit characteristics like capacitance and resistance. M_{NL} is the maximum harvested power when the EH circuit is saturated. Parameters a_{NL}, b_{NL}, and M_{NL} can be derived using a curve fitting tool [12].

To activate the tag, the harvested power should exceed the threshold, i.e., $p_k^{\mathrm{h}} \geq p_b$. In particular, $(1-\alpha_k)p_k^{\mathrm{in}} \geq p_b'$, where $p_b' \triangleq \Phi^{-1}(p_b)$ and $\Phi^{-1}(p_b) = b_{\mathrm{NL}} - \frac{1}{a_{\mathrm{NL}}}\ln\left(\frac{M_{\mathrm{NL}}-p_b}{p_b}\right)$.

III. Problem Formulation

Our objective is to optimize the BS transmit/received beamformers and the tag reflection coefficients by focusing on minimizing the total BS transmit power. The problem is thus formulated as follows:

$$\begin{aligned} (\mathrm{P1}):\ &\min_{\mathcal{A}} \quad \|\mathbf{w}\|^2 + \mathrm{Tr}(\mathbf{S}), && (13a)\\ \text{s.t}\ &\Upsilon_k \geq \Upsilon_k^{\mathrm{th}},\ \forall k, && (13b)\\ &\Gamma_u \geq \Gamma_u^{\mathrm{th}}, && (13c)\\ &\Gamma_{t,k} \geq \Gamma_k^{\mathrm{th}},\ \forall k, && (13d)\\ &p_k^{\mathrm{in}} \geq \frac{\Phi^{-1}(p_b)}{1-\alpha_k},\ \forall k, && (13e)\\ &\|\mathbf{u}_k\|^2 = 1,\ \forall k, && (13f)\\ &0 < \alpha_k < 1,\ \forall k, && (13g) \end{aligned}$$

where $\mathcal{A} = \{\{\mathbf{u}_k\}_{k\in\mathcal{K}}, \{\alpha_k\}_{k\in\mathcal{K}}, \mathbf{w}, \mathbf{S} \succ 0\}$. (13b) guarantees the sensing SINR requirement of each tag in which Υ_k^{th} denotes the targeted sensing SINR of T_k at the BS. (13c) and (13d) set the targeted SINR values for the user to decode its own data and tag data, respectively. (13e) indicates the minimum incident power required at each tag for activation. Finally, (13g) specifies the natural bounds on the reflection coefficient of each tag.

Further, combining (13c) and (13d) yield the following equivalent optimization problem:

$$\begin{aligned} (\mathrm{P2}):\ &\min_{\mathcal{A}} \quad \|\mathbf{w}\|^2 + \mathrm{Tr}(\mathbf{S}), && (14a)\\ \text{s.t}\ &\frac{|\mathbf{f}^{\mathrm{H}}\mathbf{w}|^2}{\Gamma_u^{\mathrm{th}}(1+\Gamma_k^{\mathrm{th}})} - \sum_{i\in\mathcal{K}_k}\alpha_i\left(|\mathbf{h}_i^{\mathrm{H}}\mathbf{w}|^2 + \mathbf{h}_i^{\mathrm{H}}\mathbf{S}\mathbf{h}_i\right) \geq \sigma^2, \forall k, && (14b)\\ &(13b),\ (13e) - (13g). && (14c) \end{aligned}$$

IV. Proposed Solution

Problem (14a) is non-convex due to its constraints set, which involves products of the optimization variables. Thus, we divide (14a) into three sub-problems using AO technique. For each one, we optimize (14a) for the associated variable(s) while keeping the other optimization variables fixed.

A. Sub-Problem 1: Optimization Over $\mathbf{u}_k$

For given $\{\mathbf{w}, \mathbf{S}, \{\alpha_k\}_{k\in\mathcal{K}}\}$, problem (P2) becomes a feasibility problem for receiver beamforming, $\mathbf{u}_k$. Utilizing the unique structure of the sensing SINR for each tag (10), we transform this sub-problem into a generalized Rayleigh quotient optimization problem [13]. Consequently, we obtain the following optimization problem:

$$(\mathrm{P3}):\ \max_{\mathbf{u}_k} \quad \frac{\alpha_k\mathbf{u}_k^{\mathrm{H}}\mathbf{G}_k\mathbf{R}_x\mathbf{G}_k^{\mathrm{H}}\mathbf{u}_k}{\mathbf{u}_k^{\mathrm{H}}\left(\sum_{i\in\mathcal{K}_k}\alpha_i\mathbf{G}_i\mathbf{R}_x\mathbf{G}_i^{\mathrm{H}} + \sigma_k^2\mathbf{I}_N\right)\mathbf{u}_k}, \tag{15a}$$

$$\text{s.t} \quad \|\mathbf{u}_k\|^2 = 1, \ \forall k. \tag{15b}$$

The objective function in (15a) can be restated as the following optimization problem:

$$\text{(P4)}: \ \max_{\mathbf{u}_k} \frac{\mathbf{u}_k^{\mathrm{H}} \tilde{\mathbf{G}}_k \tilde{\mathbf{G}}_k^{\mathrm{H}} \mathbf{u}_k}{\mathbf{u}_k^{\mathrm{H}} \mathbf{Q} \mathbf{u}_k}, \quad \text{s.t} \quad \|\mathbf{u}_k\|^2 = 1, \ \forall k, \tag{16a}$$

where $\tilde{\mathbf{G}}_k = \sqrt{\alpha_k}\mathbf{G}_k(\mathbf{w}+\mathbf{s})$ and $\mathbf{Q} = \sum_{i\in\mathcal{K}_k} \alpha_i \mathbf{G}_i \mathbf{R}_x \mathbf{G}_i^{\mathrm{H}} + \sigma_k^2 \mathbf{I}_N$. Problem (P4) in (16a) is a generalized Rayleigh ratio quotient problem [13]. When the transmit beamformers and reflection coefficients are fixed, the optimal received beamformer is thus given by

$$\mathbf{u}_k^* = \frac{\mathbf{Q}^{-1}\tilde{\mathbf{G}}_k}{\|\mathbf{Q}^{-1}\tilde{\mathbf{G}}_k\|}, \ \forall k, \tag{17}$$

which is an MMSE filter [13].

B. Sub-Problem 2: Optimization Over $\mathbf{w}$ *and* $\mathbf{S}$

For given $\{\mathbf{u}_k, \alpha_k\}_{k\in\mathcal{K}}$, problem (P2) can be reformulated as the following equivalent problem:

$$\text{(P5)}: \ \min_{\mathbf{w},\mathbf{S}} \quad \|\mathbf{w}\|^2 + \mathrm{Tr}(\mathbf{S}), \tag{18a}$$

$$\text{s.t} \quad \text{(14b)}, \ \text{(13b)}, \ \text{(13e)}. \tag{18b}$$

Utilizing the SDR method, we can adeptly address problem (18a) [14]. We introduce the matrix definition $\mathbf{W} = \mathbf{w}\mathbf{w}^{\mathrm{H}}$. By exploiting that $\mathbf{W}$ is semidefinite and has $\mathrm{Rank}(\mathbf{W}) = 1$, problem (18a) can be recast as (19) where the rank one constraint is dropped to relax the problem.

Note that the relaxation of the rank in (19) represents a conventional semi-definite programming (SDP) problem, which can be tackled using the CVX tool [15]. The complexity of (P5) is given by $\mathcal{O}\left(KM^3\sqrt{M}\log\left(\frac{1}{\epsilon}\right)\right)$.

C. Sub-Problem 3: Optimization Over α_k

By isolating the variables and constraints relevant to this sub-problem, we transform the original optimization problem (13a) into a feasibility problem as follows:

$$\text{(P7)}: \ \text{find} \quad \alpha_k \tag{20a}$$

$$\text{s.t} \quad \text{(13b)} - \text{(13e)}, \ \text{(13g)}, \tag{20b}$$

where any α_k that satisfies (P7) is considered a feasible solution. Following the slack variable optimization technique in [12], [14], we can introduce two new slack variables, t_1 and t_2, which represent the "SINR residual" and "EH residual" to further optimize the SINR and EH margins while satisfying constraint (20b). We then propose solving the following sub-problem:

$$\text{(P8)}: \ \min_{\alpha_k, t_1, t_2} \quad \lambda_1 t_1 + \lambda_2 t_2 \tag{21a}$$

$$\text{s.t} \quad \alpha_k \mathbf{u}_k^{\mathrm{H}} \mathbf{G}_k \mathbf{R}_x \mathbf{G}_k^{\mathrm{H}} \mathbf{u}_k \geq \tag{21b}$$

$$\Upsilon_k^{\mathrm{th}} \mathbf{u}_k^{\mathrm{H}} \left(\sum_{i\in\mathcal{K}_k} \alpha_i \mathbf{G}_i \mathbf{R}_x \mathbf{G}_i^{\mathrm{H}} + \sigma_k^2 \mathbf{I}_N \right) \mathbf{u}_k + t_1, \forall k,$$

$$\frac{|\mathbf{f}^{\mathrm{H}}\mathbf{w}|^2}{\Gamma_u^{\mathrm{th}}(1+\Gamma_k^{\mathrm{th}})} - \sum_{i\in\mathcal{K}_k} \alpha_i \left(|\mathbf{h}_i^{\mathrm{H}}\mathbf{w}|^2 + \mathbf{h}_i^{\mathrm{H}}\mathbf{S}\mathbf{h}_i\right) \geq \sigma^2, \forall k, \tag{21c}$$

$$(1-\alpha_k)p_k^{\mathrm{in}} \geq \Phi^{-1}(p_b) + t_2, \forall k, \tag{21d}$$

$$\text{(13g)}, \tag{21e}$$

where λ_1 and λ_2 are positive constants. Problem (21a) is convex and thus can be efficiently solved by solvers such as CVX [15]. Our algorithm to solve (13a) is presented in Algorithm 1. Algorithm 1 iterations yield a non-increasing sequence of objective values with guaranteed convergence. The proof follows similar steps as [16].

Algorithm 1 AO Algorithm

Input: Set $t = 0$, the convergence tolerance $\epsilon > 0$, initial solution $\mathcal{A}^0$ and objective function value $F^{(0)} = 0$.
while $\frac{F^{(t+1)} - F^{(t)}}{F^{(t+1)}} \geq \epsilon$ **do**
  Solve (17) for the received beamformer, $\mathbf{u}_k^{(t+1)}$.
  Solve (18a) to obtain $\{\mathbf{w}^{(t+1)}, \ \mathbf{s}^{(t+1)}\}$ by recovering a rank-one solution via Gaussian randomization
  Solve (21a) for the reflection coefficients, $\alpha_k^{(t+1)}$.
  Calculate the objective function value $F^{(t+1)}$.
  Set $t \leftarrow t+1$;
end while
Output: Optimal solutions $\mathcal{A}^*$.

TABLE I: Simulation parameters.

Parameter	Value	Parameter	Value
f_c	3 GHz	Γ_u^{th}	1 bps/Hz
B	10 MHz	Γ_k^{th}	1 bps/Hz
N_f	10 dB	p_b	-20 dBm
$M = N$	8	M_{NL}	20×10^{-3} W
K	3	a_{NL}	6400
Υ_k^{th}	1 bps/Hz	b_{NL}	0.003

V. Simulation Results

We next present simulation results for assessing the performances of the proposed ISABC network and the AO algorithm.

A. Simulation Setup and Parameters

The 3GPP urban micro (UMi) model is adopted to model the path-loss $\{\zeta_a, \zeta_b\}$ with $f_c = 3$ GHz operating frequency [17, Table B.1.2.1]. We model the AWGN variance, σ^2, as $\sigma^2 = 10\log_{10}(N_0 B N_f)$ dBm, where $N_0 = -174$ dBm/Hz, B is the bandwidth, and N_f is the noise figure. Unless otherwise specified, Table I summarizes the simulation parameters. All simulations are evaluated for 10^3 iterations. Considering a smart home use case, we place the BS and the user at $\{0, 0\}$ and $\{12, 0\}$, respectively, while the tags are randomly distributed within a circle centered at $\{6, -4\}$ with a radius of 3 m.

In evaluating the proposed ISABC system with passive backscatter tags (ISABC-P), we compare it against key benchmarks for a comprehensive analysis. These include the conventional ISAC system with an FD BS and traditional radar targets, a basic BackCom setup without BS sensing, a communication-focused scheme in a single-user environment, and a sensing-focused approach excluding primary communication. Additionally, we assess the system with active tags (ISABC-A) to observe the impact of removing EH

$$
\begin{aligned}
(\text{P6}): \ & \underset{\mathbf{W},\mathbf{S}}{\text{minimize}} \ \ \text{Tr}(\mathbf{W}) + \text{Tr}(\mathbf{S}) \\
\text{s.t} \ \ & \Upsilon_k^{\text{th}} \mathbf{u}_k^{\text{H}} \left(\sum_{i \in \mathcal{K}_k} \alpha_i \left(\text{Tr}(\mathbf{G}_i^{\text{H}} \mathbf{G}_i \mathbf{W}) + \text{Tr}(\mathbf{G}_i^{\text{H}} \mathbf{G}_i \mathbf{S}) \right) + \sigma_k^2 \mathbf{I}_N \right) \mathbf{u}_k - \alpha_k \mathbf{u}_k^{\text{H}} \left(\text{Tr}(\mathbf{G}_i^{\text{H}} \mathbf{G}_i \mathbf{W}) + \text{Tr}(\mathbf{G}_i^{\text{H}} \mathbf{G}_i \mathbf{S}) \right) \mathbf{u}_k \leq 0, \forall k, \\
& \frac{\text{Tr}(\mathbf{f}\mathbf{f}^{\text{H}} \mathbf{W})}{\Gamma_u^{\text{th}} (1 + \Gamma_k^{\text{th}})} - \sum_{i \in \mathcal{K}_k} \alpha_i \left(\text{Tr}(\mathbf{h}_i \mathbf{h}_i^{\text{H}} \mathbf{W}) + \text{Tr}(\mathbf{h}_i \mathbf{h}_i^{\text{H}} \mathbf{S}) \right) \geq \sigma^2, \forall k, \\
& P_{\text{th}} - (1 - \alpha_k)(\text{Tr}(\mathbf{g}_{f,k} \mathbf{g}_{f,k}^{\text{H}} \mathbf{W}) + \text{Tr}(\mathbf{g}_{f,k}^{\text{H}} \mathbf{g}_{f,k} \mathbf{S})) \leq 0, \forall k.
\end{aligned} \tag{19}
$$

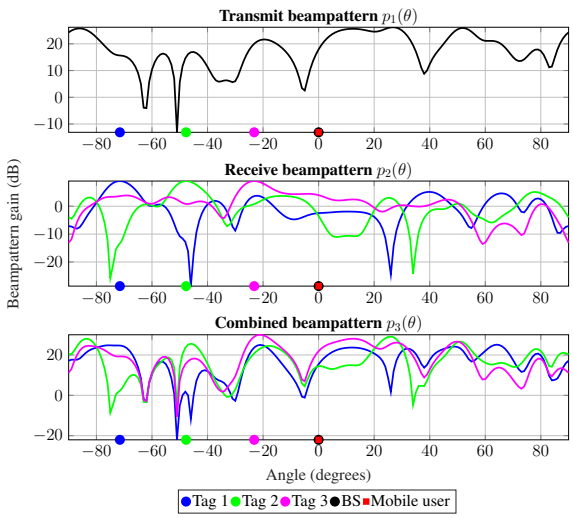


Fig. 2: Beampattern regarding radar functionality of Algorithm 1.

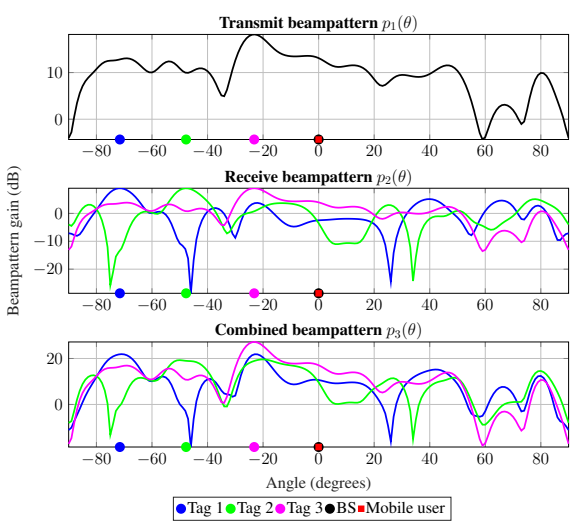


Fig. 3: Beampattern regarding communication functionality of only communication.

requirements, alongside scenarios with random tag reflection coefficients, and the performance implications of MF and ZF beamformers. The MF and ZF beamformers are given by $\mathbf{u}_{\text{MF}} = \mathbf{H}_b$ and $\mathbf{u}_{\text{ZF}} = \mathbf{H}_b \left(\mathbf{H}_b^{\text{H}} \mathbf{H}_b \right)^{-1}$, respectively, where $\mathbf{H}_b \in \mathbb{C}^{N \times K}$ symbolizes the backward channel matrix and each k-th column's vector is defined as $\mathbf{g}_{b,k}$ for $k \in \mathcal{K}$.

B. Beampattern Gains

Modern radar functionality has evolved to harness the power of beamforming, i.e., directly transmit and receive beams in specific directions. Algorithm 1 serves this function, guiding the formation and steering of these beams. Beamforming in radar systems converges signals from an array of antennas, crafting a directed "beam" or "lobe" [5], [18]. Intriguingly, this beam can be electronically steered while the antennas remain stationary. This electronic steering capability amplifies signal quality, boosts backscatter tag detection, and significantly minimizes potential interference [5], [18].

The transmit signal, representing the outward-projected energy, is critical in efficiently illuminating the radar's targets. Meanwhile, the received beamformer, normalized as $\|\mathbf{u}_k^*\| = 1$, is optimized for clear reception. Three pivotal beampatterns arise from these components:

$$p_1(\theta) = \left| \mathbf{b}^{\text{H}}(\theta_k) \mathbf{x}^* \right|^2, \tag{22a}$$

$$p_2(\theta) = \left| (\mathbf{u}_k^*)^{\text{H}} \mathbf{b}(\theta_k) \right|^2, \tag{22b}$$

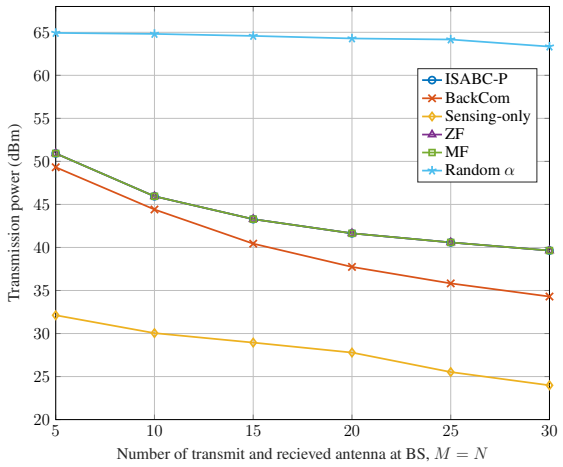
$$p_3(\theta) = \left| (\mathbf{u}_k^*)^{\text{H}} \mathbf{b}(\theta_k) \mathbf{b}^{\text{H}}(\theta_k) \mathbf{x}^* \right|^2. \tag{22c}$$

First, (22a) illustrates how the transmitted energy disperses as a function of angle θ. Second, (22b) encapsulates the sensitivity of the radar system across different angles during the reception of reflected energy. Finally, (22c) offers a combined representation, integrating the effects of transmission and subsequent reflection processing.

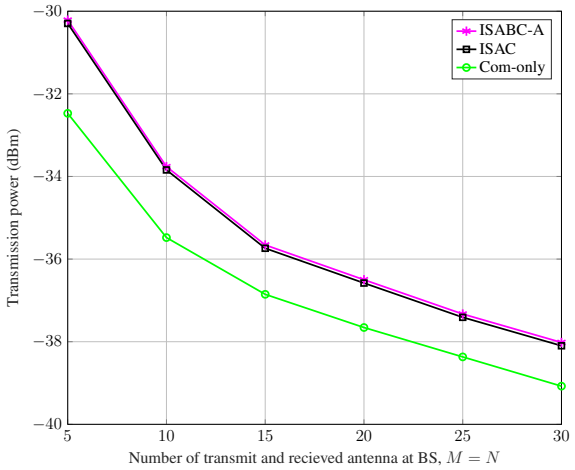
Fig. 2 and Fig. 3 make it easier to discern the nuances and effectiveness of beamforming algorithms.

Fig. 4a and Fig. 4b show the BS transmit power requirement as a function of $M = N$. As per Fig. 4a, the random-α benchmark is the most inefficient regarding energy use. The sensing-only approach requires less transmit power than ISABC-P, as it eliminates the communication performance. Furthermore, sub-optimal MRT- and ZF-based beamformers closely align with our proposed schemes but need CSI. Importantly, ISABC-P allows for sensing with low-cost tags, an essential feature for IoT networks, with only a slight increase in transmit power. For instance, a $3.4\,\%$ increase in transmit power with $M = N = 10$ results in a $75\,\%$ sum rate gain, i.e., user rate + tags' rate + sensing rate.

Fig. 4b reveals that ISAC and purely communication-focused methods require less BS transmit power than our ISABC-P. The latter needs more BS power to deliver sufficient power at the tags for EH. Of course, this is the cost of using



(a)



(b)

Fig. 4: Transmit power versus the number of antennas at the BS, $M = N$, for different schemes.

fully passive tags. Nevertheless, utilizing active tags, as in ISABC-A, can obviate the need for this additional power at the BS. For instance, with $M = N = 10$, ISABC-A only requires a $0.24\,\%$ increase in transmit power to provide a $75\,\%$ sum rate gain over conventional ISAC. On the other hand, active tags with complex tag designs are more expensive than passive tags, and the type of tag used may vary depending on the application.

VI. Conclusion

In this study, we introduced the concept of ISABC, an innovative system that combines BackCom and sensing with an FD BS. The FD BS plays a dual role, serving as a sensor through backscatter tag detection and enabling user communication. Using the AO technique, we derived precise communication (both primary and backscatter) and sensing rates, focusing on minimizing power consumption at the BS. This approach empowers passive and active tags for sensing and communication, opening doors to new possibilities. Future research can explore advanced modulation and machine learning to enhance ISABC's adaptability in urban environments, aligning with IoT and 6G technologies for comprehensive networks.

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